OCR Further Mathematics
Pure Core - Further Calculus
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Exam Papers Practice

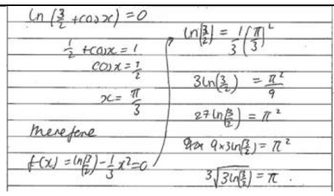
Question			Answer/Indicative content	Marks	Part marks and guidance	
1			For AB, $V = \pi \times 1^2 \times 4$ 12.566… () For BC, $V = \int_a^b \pi x^2 \mathrm{d}y = \pi\int_4^9 \left(37-(y-10)^2\right)\mathrm{d}y$ 356.05… $\Rightarrow$ Total V = 356.05… + 12.566… = 368.61… = 369 (cm^3) ti 3 sf	M1 (AO 3.3) A1 (AO 1.1) A1 (AO 3.4) [3]	4π Split into two parts and use formulae An integral and an attempt at the volume of a cylinder must be seen Integration – ignore limits BC $\frac{340}{3}\pi$ Units are not required $\frac{352}{3}\pi$	
			Total	3		
2	a	i	$\mathrm{f}(0) = \frac{\pi}{4}$	B1 (AO 1.1) [1]	Not for 45^0	
		ii	$\mathrm{f}'(x) = \frac{1}{1+(1+x)^2} \Rightarrow \mathrm{f}'(0) = \frac{1}{2}$	M1 (AO 2.1) A1 (AO 1.1) [2]	Diffn – Must be seen $\mathrm{f}'(x) = \frac{1}{1+x^2}$ is M0	
		iii	$\mathrm{f}'(x) = \frac{1}{1+(1+x)^2} = \frac{1}{2+2x+x^2}$ $\Rightarrow \mathrm{f}''(x) = \frac{1}{(2+2x+x^2)^2} \times (-1) \times (2+2x)$ $= \frac{-(2+2x)}{(2+2x+x^2)^2}$ $\Rightarrow \mathrm{f}''(0) = \left(\frac{-2}{4}\right) = -\frac{1}{2}$	M1 (AO 2.1) A1 (AO 2.1) A1 (AO 2.1) [3]	Diffn *their* f'(x) oe, e.g. $\mathrm{f}''(x) = -\frac{2(1+x)}{\left(1+(1+x)^2\right)^2}$ f''(0) must be seen. The substitution must be seen (implied by $-\frac{2}{4}$) AG	
	b		$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \mathrm{f}''(0)\frac{x^2}{2}$ $= \frac{\pi}{4} + \frac{1}{2}x - \frac{1}{2} \times \frac{x^2}{2}$ $= \frac{\pi}{4} + \frac{x}{2} - \frac{x^2}{4}$	M1 (AO 1.1) A1 (AO 2.2a) [2]	Using the formula and substituting *their* value for f;(0) ft their values from above part	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	8	

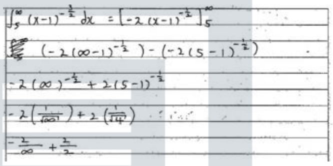
Question			Answer/Indicative content	Marks	Part marks and guidance	
3	a		DR $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ... = (1 + x) + \left(\frac{x^2}{2!} + \frac{x^3}{3!} + ...\right)$ $x > 0 \Rightarrow \frac{x^2}{2!} + \frac{x^3}{3!} + ... > 0$ $\Rightarrow e^x > 1 + x$	M1 (AO1.1) A1 (AO2.2a) [2]	Quoting and *using* the Maclaurin series AG. Result with sufficient justification	
	b		DR $t = x + 1 \Rightarrow e^{t-1} > t \Rightarrow \frac{e^t}{e} > t \Rightarrow e^t > et$	B1 (AO3.1a) [1]	AG	
	c		DR $t = \frac{\pi}{e} > 1$ since $2 < e < 3$ and $\pi > 3$ $e^{\frac{\pi}{e}} > e \times \frac{\pi}{e} (= \pi)$ $\Rightarrow e^{\pi} > \pi^e$ (ie e^{π} is greater) Alternative method $t = \ln \pi$ $e^{\ln \pi} > e \ln \pi$ $e^{\ln \pi} > e \ln \pi$ $\pi > \ln(\pi^e)$ $e^{\pi} > \pi^e$	B1 (AO3.1a) M1 (AO3.1a) A1 (AO1.1) Alternative method B1 M1 A1 [3]	Some justification that $t > 1$ is required Substituting their choice into the inequality Answer without use of inequality in part above scores M0A0 Alternative method Some justification that $t > 1$ is required	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4	a		$f(0) = \ln(\frac{1}{2} + \cos 0) = \ln(\frac{3}{2})$ $\frac{d}{dx}\left(\ln(\frac{1}{2} + \cos x)\right) = \frac{-\sin x}{\frac{1}{2} + \cos x} \Rightarrow f'(0) = 0$ $\frac{d^2 \ln\left(\frac{1}{2}+\cos x\right)}{dx^2} = \frac{-\cos x\left(\frac{1}{2}+\cos x\right)+\sin x(-\sin x)}{\left(\frac{1}{2}+\cos x\right)^2}$ $\ldots \Rightarrow f''(0) = -\frac{2}{3}$ $\ln\left(\frac{1}{2}+\cos x\right) = \ln\left(\frac{3}{2}\right) - \frac{x^2}{3} + \ldots$	B1(AO1.1) M1(AO3.1a) A1(AO1.1) A1(AO1.1)[4]	soi Differentiating using chain rule (or rule for ln(f(x)) and evaluating when $x = 0$ Differentiating again using quotient (or product / chain) rule. www	Allow sign error in numerator NB Simplifies to $-\frac{\frac{1}{2}\cos x + 1}{\left(\frac{1}{2}+\cos x\right)^2}$ If zero scored then SC1 for correct expansion
					<u>Examiner's Comments</u> Most candidates correctly found f(0), however sign errors often crept into their expressions for f'(x) or f''(x). A few candidates ignored the requirement to 'use differentiation' and attempted (usually unsuccessfully) to find the terms via the use of standard formulae booklet expansions. Where they did find the correct expansion they were given partial credit.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		$\ln\left(\frac{1}{2}+\cos x\right)=0 \Rightarrow x=\frac{\pi}{3}$ (or $-\frac{\pi}{3}$) $\therefore \ln\left(\frac{3}{2}\right)-\frac{\left(\frac{\pi}{3}\right)^2}{3}\approx 0$ $\ln\left(\frac{3}{2}\right)-\frac{\pi^2}{27}\approx 0 \Rightarrow \pi \approx \sqrt{27\ln\left(\frac{3}{2}\right)}=3\sqrt{3\ln\left(\frac{3}{2}\right)}$	B1(AO1.1)M1(AO3.1a)A1(AO3.2a)[3]	Finding either $\pm\pi/3$ as a root. Allow 60° for B1. Ignore other roots Substituting their root, in radians, into their Maclaurin series and equating (approximately) to 0. Could see ± but must be removed by final conclusion. Must use approximately equals symbol (not just equals symbol)	Or equating their expression (approximately) to 0 and rearranging for x: $\ln\left(\frac{3}{2}\right)-\frac{x^2}{3}\approx 0 \Rightarrow x\approx\sqrt{3\ln\left(\frac{3}{2}\right)}$ $\frac{\pi}{3}\approx\sqrt{3\ln\left(\frac{3}{2}\right)} \Rightarrow \pi\approx 3\sqrt{3\ln\left(\frac{3}{2}\right)}$	
					<u>Examiner's Comments</u> Some candidates considered only the original equation and others only the modified equation with Maclaurin expansion. Those who considered both were usually able to put the two together for full method marks. Some otherwise good answers were marred by totally ignoring the 'approximately equal' sign. Exemplar 5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					 $\ln\left(\frac{3}{2}+\cos x\right)=0$ $\frac{1}{2}+\cos x = 1$ $\cos x=\frac{1}{2}$ $x=\frac{\pi}{3}$ therefore $f(x)=\ln\left(\frac{3}{2}\right)-\frac{1}{3}x^2=0$ $\ln\left(\frac{3}{2}\right)=\frac{1}{3}\left(\frac{\pi}{3}\right)^2$ $3\ln\left(\frac{3}{2}\right)=\frac{\pi^2}{9}$ $27\ln\left(\frac{3}{2}\right)=\pi^2$ $9\times 3\ln\left(\frac{3}{2}\right)=\pi^2$ $3\sqrt{3\ln\left(\frac{3}{2}\right)}=\pi$ This candidate appears unaware of the significance of the approximately equals sign in the question and / or of the approximation involved in using only the first few terms of a Maclaurin expansion.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5			DR $\int (x-1)^{-\frac{3}{2}}\mathrm{d}x = -2(x-1)^{-\frac{1}{2}}(+c)$ $\int_5^N (x-1)^{-\frac{3}{2}}\mathrm{d}x = \left[-2(x-1)^{-\frac{1}{2}}\right]_5^N$ $-\dfrac{2}{\sqrt{N-1}}+\dfrac{2}{\sqrt{5-1}}$ $\lim\limits_{N\to\infty}\dfrac{1}{\sqrt{N-1}}=0$ oe $\int_5^\infty (x-1)^{-\frac{3}{2}}\mathrm{d}x = \lim\limits_{N\to\infty}\left\{-\dfrac{2}{\sqrt{N-1}}+\dfrac{2}{\sqrt{5-1}}\right\}=1$	B1(AO1.1a)M1(AO2.1)A1(AO1.1)B1(AO2.1)A1(AO2.2a) [5]	Consideration of a finite upper limit Not *just* eg $\dfrac{1}{\infty}=0$ AG. Convincing argument equating improper integral to solution	Can be seen as part of limit of both terms, but must be explicitly shown as zero
					Examiner's Comments This question requested detailed reasoning. Good responses to this problem were elegant and showed a real understanding of the process. Usually these candidates could string together their whole argument via a series of . They then justified their ualities, starting with evaluation of the limit with an aside on the right hand side of their page at the relevant place in the argument. Many candidates, however, simply did not approach this problem with sufficient rigour, for instance · treating infinity as a number (a common starting point for candidates was 'Let $a = \infty$' which immediately cut down the number of marks available), or · not presenting the full	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					argument or not even mentioning the entity whose value was supposed to be being shown to be equal to 1, or • using poor, confusing or ambiguous notation. As mentioned in Q1(b) there seems to be quite a lot of confusion around the whole issue of limits. For example, candidates should realise that a limit is a value and so cannot, in itself, tend to anything. Exemplar 1  $\int_5^\infty (x-1)^{-\frac{3}{2}}\,dx = \left[-2(x-1)^{-\frac{1}{2}}\right]_5^\infty$ $(-2(\infty-1)^{-\frac{1}{2}}) - (-2(5-1)^{-\frac{1}{2}})$ $-2(\infty)^{-\frac{1}{2}} + 2(5-1)^{-\frac{1}{2}}$ $-2\left(\frac{1}{\sqrt{\infty}}\right) + 2\left(\frac{1}{\sqrt{4}}\right)$ $-\frac{2}{\infty} + \frac{2}{2}$ Because this candidate doesn't use a limiting process applied to a proper integral, they cannot access most of the marks available.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance		
6	a	i	$f'(x)=\frac{1}{(1-x^2)^{\frac{1}{2}}}$ from the formula book so $f''(x)=-\frac{1}{2}.\frac{1}{(1-x^2)^{\frac{3}{2}}}.(-2x)$ $=\frac{x}{(1-x^2)^{\frac{3}{2}}}$	M1 (AO1.1) A1 (AO1.1) [2]	Formula from the Formula Booklet and attempt differentiation	To within sign error	
		ii	$f(0) = 0$, $f'(0) = 1$ and $f''(0) = 0$ $f'''(x)=\frac{(1-x^2)^{\frac{3}{2}}-x.\frac{3}{2}(1-x^2)^{\frac{1}{2}}.(-2x)}{(1-x^2)^3}$ so $f'''(0) = 1$ and $f(x)=x+\frac{1}{6}x^3+...$	B1 (AO1.1) M1 (AO3.1a) A1 (AO2.1) [3]	or $a_0 = 0$, $a_1 = 1$ and $a_2 = 0$ Differentiate and simplify far enough to be able to justify value 1 Condone 3! In place of 6	Ignore sign error in $f''(x)$ Either full derivative or "zero term" denoted as such Not BC. If M0 then SC1 for correct expansion	
		iii	$\int_0^{\frac{1}{2}} f(x)\,dx \approx \int_0^{\frac{1}{2}} x+\frac{1}{6}x^3\,dx$ $= 0.127604167...$ $= 0.127604$ to 6 dp	M1 (AO1.1) A1 (AO1.1) [2]	Integral of their 2 term cubic with limits Could be BC		

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		$\int 1\times\sin^{-1}x\,dx = x\sin^{-1}x - \int \frac{x}{\sqrt{1-x^2}}dx$ $= x\sin^{-1}x + (1-x^2)^{\frac{1}{2}} \quad (+c)$ $\int_0^{\frac{1}{2}} f(x) = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) [3]	Attempt integration by parts Examiner's Comments In (a)(ii) the command word "determine" notifies candidates that they cannot merely use their calculator to find f'''(0). Of course, the calculator could be used here (and in questions such as Question 1) to double check responses and alert candidates as to errors they may have made. In (b) while most candidates were able to gain the first mark by attempting integration by parts, several were unable to tackle the integral of $\frac{x}{(1-x^2)^{\frac{1}{2}}}$. Centres may find it helpful to allow plenty of practice of selecting a suitable method of integration when facing such expressions. Candidates should then be alert to the significance, here, of 'x' being the derivative of 'x^2' (to within a constant term).	ignore limits. Formula for parts must be correct
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7			$\frac{4}{1-x^4}=\frac{A}{1-x}+\frac{B}{1+x}+\frac{Cx+D}{1+x^2}$			
			$\Rightarrow A(1+x)(1+x^2)+B(1-x)(1+x^2)+(Cx+D)(1-x^2)=4$	M1 (AO3.1a)	Proper split to produce integrable integrand	
			$x=1: 4A=4 \Rightarrow A=1$	A1 (AO1.1)	For one of the terms / constants	
			$x=-: 4B=4 \Rightarrow B=1$ $x=0: A+B+D=4 \Rightarrow D=2$ $x^3: A-B-C=0 \Rightarrow C=0$	A1 (AO1.1)	For all terms / constants Or $\frac{1}{4}$ of these	
			$I=\int_0^{\frac{1}{\sqrt{3}}}\left(\frac{1}{1-x}+\frac{1}{1+x}+\frac{2}{1+x^2}\right)\mathrm{d}x$		See below for other possibilities	
			$=\left[\ln\left(\frac{1+x}{1-x}\right)+2\tan^{-1}x\right]_0^{\frac{1}{\sqrt{3}}}$	M1 (AO1.1)	Correctly integration of their integrand without simplification – ignore limits	
			$=\ln\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)+2\tan^{-1}\frac{1}{\sqrt{3}}\quad(-0)=\ln\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)+\frac{\pi}{3}$	M1 (AO1.1)	Substitution – ignore – 0	
			$=\ln\left(2+\sqrt{3}\right)+\frac{\pi}{3}$ i.e. $a=2$, $b=3$, $c=3$	A1 (AO1.1)		
				M0	Values must be stated	
			$\frac{4}{1-x^4}=\frac{A}{1-x}+\frac{B}{1+x}+\frac{D}{1+x^2}$		But consider last 3 marks for correct integration	
			$\frac{4}{1-x^4}=\frac{A}{1-x^2}+\frac{D}{1+x^2}$ with A = D = 2 by inspection	M1 A1 A1		
			Followed by $\frac{2}{1+x^2}+\frac{1}{1-x}+\frac{1}{1+x}$ in integration section		Look to see the second split further on in question	
			$\frac{4}{1-x^4}=\frac{A}{1-x^2}+\frac{D}{1+x^2}$	M1 A1 A1		
			$\Rightarrow\int_0^{\frac{1}{\sqrt{3}}}\frac{4}{1-x^4}\mathrm{d}x=\int_0^{\frac{1}{\sqrt{3}}}\left(\frac{2}{1-x^2}+\frac{2}{1+x^2}\right)\mathrm{d}x$	[6]	Look for the integration. If $\tanh^{-1}$ is used then give full marks here.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$=\left[2\tanh^{-1}x+2\tan^{-1}x\right]_0^{\frac{1}{\sqrt{3}}}=\left(2\tanh^{-1}\frac{1}{\sqrt{3}}+2\tan^{-1}\frac{1}{\sqrt{3}}\right)$ $=\ln\left(\frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}}\right)+2\frac{\pi}{6}-0$ $=\ln\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)+\frac{\pi}{3}=\ln\left(\frac{1+2\sqrt{3}+3}{2}\right)+\frac{\pi}{3}=\ln\left(\frac{4+2\sqrt{3}}{2}\right)+\frac{\pi}{3}$ $=\ln\left(2+\sqrt{3}\right)+\frac{\pi}{3}$		<u>Examiner's Comments</u> Many candidates appreciated that the integrand had to be split into partial fractions in order to perform the integration. The $\frac{D}{1+x^2}$ instea $\frac{Cx+D}{1+x^2}$ inclusion d of of was not a correct split and so did not earn the first 3 marks. However, since C = 0 the result was the same and so the last 3 marks could be earned. The $\frac{4}{1-x^4}=\frac{2}{1+x^2}+\frac{2}{1-x^2}$ wa result s only acceptable in two circumstances. (i) If $\frac{2}{1-x^2}$ was split further to give $\frac{2}{1-x^2}=\frac{1}{1+x}+\frac{1}{1-x}$ as a further process during integration. (ii) $\frac{2}{1-x^2}$ was seen to give $\tanh^{-1}x$ when integrated.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8	a		$2\sinh^2 u + 1 \equiv 2\left(\frac{e^u - e^{-u}}{2}\right)^2 + 1$	M1 (AO2.1)	Use of exponential form for sinh u	
			$= \frac{e^{2u} - 2 + e^{-2u}}{2} + 1 \equiv \frac{e^{2u} + e^{-2u}}{2} \equiv \cosh 2u$ AG	A1 (AO2.1) [2]		
	b		$\cosh 2u \equiv 2\sinh^2 u + 1$ $\Rightarrow 2\sinh 2u \equiv 4\sinh u \cosh u$ $\Rightarrow \sinh 2u \equiv \sinh u \cosh u$ AG	B1 (AO2.1) [1]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		$x = \sinh^2 u \Rightarrow dx = 2\sinh u \cosh u\, du$	M1 (AO3.1a)	Attempt to find $\frac{dx}{du}$	
			$\Rightarrow \int\sqrt{\frac{x}{x+1}}dx = \int\sqrt{\frac{\sinh^2 u}{\sinh^2 u+1}}2\sinh u\cosh u\ du$			
			$= 2\int \sinh^2 u\, du$	A1 (AO1.1)		
			$= \int (\cosh 2u - 1)\, du$	M1 (AO1.1a)	Use double angle formulae and attempt to integrate.	
			$= \frac{1}{2}\sinh 2u - u + c = \sinh u\cosh u - u + c$	A1 (AO1.1)	Ignore c.	
			$= \sqrt{x(1+x)} - \sinh^{-1}\sqrt{x} + c$			
			So $f(x) = \sqrt{x(1+x)} + c$, $a = -1$, $b = 1$	A1 (AO1.1)	c must be included here as part of f(x) – allow a and b not being stated explicitly but f(x) must be	
			Alternative method			
			$= 2\int\left(\frac{e^u - e^{-u}}{2}\right)^2 du = \frac{1}{2}\int e^{2u} - 2 + e^{-2u}\ du$	M1 (AO1.1a)	Use exponentials and attempt to integrate.	
			$= \frac{1}{4}e^{2u} - \frac{1}{4}e^{-2u} - u + c = \frac{1}{2}\sinh 2u - u + c$	A1 (AO1.1)	Ignore c.	
			$= \sqrt{x(1+x)} - \sinh^{-1}\sqrt{x} + c$			
			So $f(x) = \sqrt{x(1+x)} + c$, $a = -1$, $b = 1$	A1 (AO1.1) [5]	c must be included here as part of f(x) – allow a and b not being stated explicitly but f(x) must be	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		$\text{Area} = \left[\sqrt{x(1+x)} - \sinh^{-1}\sqrt{x}\right]_1^2$ $= \left(\sqrt{6} - \ln(\sqrt{2}+\sqrt{3})\right) - \left(\sqrt{2} - \ln(1+\sqrt{2})\right)$ $= \sqrt{6} - \sqrt{2} + \ln\left(\dfrac{1+\sqrt{2}}{\sqrt{2}+\sqrt{3}}\right)$ So $p = \sqrt{6} - \sqrt{2}$, $q =$ 1, $r = \dfrac{1+\sqrt{2}}{\sqrt{2}+\sqrt{3}}$	M1 (AO1.1) A1 (AO1.1) [2]	Correct limits substituted and subtracted into *their* answer to (c) soi *p, q, r* must be stated <u>Examiner's Comments</u> Part (a) was standard but a number of candidates found an identity in exponentials without mentioning or cosh 2*u*. or sinh*u*. In part (c) many did not do a complete substitution, ending with an integral in *u* to be integrated with respect to *x*. Those that progressed well here, recognising also that the result of part (a) was required, often did not complete the question to give the response in the form stated. Unless part (c) was correct, the response for part (d) would not take the form given, although a substitution into their response for part (c) gained the method mark.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9			$\text{Mean value} = \frac{1}{3}\int_0^3 f(x)\mathrm{d}x$ $= \frac{1}{3}\int_0^3 x^2 + 6x\mathrm{d}x = 12$	M1 (AO1.1) A1 (AO1.1) [2]	Use the correct formula BC <u>Examiner's Comments</u> The majority of candidates knew and could apply the formula for the mean value and scored both marks.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance		
10	a		$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{e^{2x}+1}{2e^x}$ $\Rightarrow \operatorname{sech} x = \frac{2e^x}{e^{2x}+1}$ AG	M1(AO1.1) A1(AO2.1) [2]	Use of cosh$x$ in exponentials <u>Examiner's Comments</u> This was done well by almost all candidates.		
	b		$u = e^x \Rightarrow du = e^x\,dx$ $\Rightarrow dx = \frac{du}{u}$ $\Rightarrow \int \operatorname{sech} x\,dx = \int\left(\frac{2e^x}{e^{2x}+1}\right)dx$ $= \int \frac{2u}{u^2+1}.\frac{du}{u}$ = 2 $\tan^{-1}(u)$ + c = 2 $\tan^{-1}(e^x)$ + c Alternatively: u = sinh$x$ ⇒ du = cosh$x$dx M1 $\Rightarrow \int \operatorname{sech} x\,dx = \int \frac{1}{\cosh x}.\frac{du}{\cosh x} = \int \frac{du}{\cosh^2 x}$ $= \int \frac{du}{1+\sinh^2 x} = = \int \frac{du}{1+u^2}$ A1 = $\tan^{-1} u + c$ M1 = $\tan^{-1}(\sinh x) + c$ A1 Alternatively: $\int \operatorname{sech} x\,dx = \int \frac{2e^x}{e^{2x}+1}dx$ Let $e^x = \tan u \Rightarrow e^x dx = \sec^2 u\,du \Rightarrow dx = \frac{\sec^2 u}{\tan u}du$	M1(AO3.1a) A1(AO1.1) M1(AO3.1a) A1(AO1.1) [4]	Substitute and use (a) any form entirely in terms of u Use standard form for integral and substitute back Must include c	Allow absence of du	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			M1 $\Rightarrow \int \text{sech}x\,dx =$ $\int \frac{2\tan u}{\tan^2 u+1} \cdot \frac{\sec^2 u}{\tan u} du = 2\int du$ A1 $2u + c$ M1 $2\tan^{-1}(e^x) + c$ A1		Substitute In correct form Integrate and substitute back Must include c <u>Examiner's Comments</u> Most candidates made a good choice of substitution to obtain the indefinite integral. A number of candidates however lost the final accuracy mark by not including an arbitrary constant of integration.		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
11	a		$=\dfrac{1}{1+\left(\sqrt{2x}\right)^2}\times\dfrac{d}{dx}\left(\sqrt{2x}\right)$ oe $=\dfrac{1}{1+2x}\times\dfrac{\sqrt{2}}{2\sqrt{x}}=\dfrac{1}{1+2x}\times\dfrac{1}{\sqrt{2x}}$ Alternatively: $\tan y=\sqrt{2x}$ $\Rightarrow \sec^2 y\dfrac{dy}{dx}=\dfrac{\sqrt{2}}{2\sqrt{x}}\Rightarrow\left(1+\tan^2 x\right)\dfrac{dy}{dx}=\dfrac{\sqrt{2}}{2\sqrt{x}}$ M1 $\Rightarrow\dfrac{dy}{dx}=\dfrac{1}{1+2x}\times\dfrac{\sqrt{2}}{2\sqrt{x}}$ A1	M1(AO1.1a) A1(AO1.1) [2] [4]	Attempt to differentiate using chain rule Make a substitution **Examiner's Comments** It was best to use the chain rue to perform this differentiation, although a significant number did not do so. The alternative (to make a substitution) was usually correct, although it took more writing (and therefore time) to do so.	i.e. product of 2 terms	

Question	Answer/Indicative content	Marks	Part marks and guidance
b	$\int_{1/6}^{1/2} \frac{1}{(1+2x)\sqrt{x}}\mathrm{d}x = \sqrt{2}\int_{1/6}^{1/2} \frac{1}{(1+2x)\sqrt{2x}}\mathrm{d}x$ $= \sqrt{2}\left[\tan^{-1}\sqrt{2x}\right]_{1/6}^{1/2} = \sqrt{2}\left(\tan^{-1}1 - \tan^{-1}\frac{1}{\sqrt{3}}\right)$ $= \sqrt{2}\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \frac{\sqrt{2}}{12}\pi$ So $k = \frac{\sqrt{2}}{12}$ Alternatively: Let $u = \sqrt{x}$ M1 $\mathrm{d}u = \frac{1}{2\sqrt{x}}\mathrm{d}x \Rightarrow \mathrm{d}x = 2\sqrt{x}\mathrm{d}u = 2u\mathrm{d}u$ $\int_{\frac{1}{6}}^{\frac{1}{2}} \frac{\sqrt{x}}{\left(x+2x^2\right)}\mathrm{d}x = \int_{x=\frac{1}{6}}^{x=\frac{1}{2}} \frac{u}{\left(u^2+2u^4\right)}2u\mathrm{d}u = 2\int_{x=\frac{1}{6}}^{x=\frac{1}{2}} \frac{1}{\left(1+2u^2\right)}\mathrm{d}u$ A1 $= \sqrt{2}\left[\tan^{-1}u\sqrt{2}\right]_{x=\frac{1}{6}}^{x=\frac{1}{2}}$ M1 $= \sqrt{2}\left[\tan^{-1}\sqrt{2x}\right]_{x=\frac{1}{6}}^{x=\frac{1}{2}} = \sqrt{2}\left(\tan^{-1}1 - \tan^{-1}\frac{1}{\sqrt{3}}\right) = \sqrt{2}\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$ $= \frac{\pi\sqrt{2}}{12}$ A1	M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1(AO1.1) [4]	Get into form of (a). Ignore limits Correct form Use (a) and correct limits in correct order. oe Make a substitution Get into correct form Use standard result with correct limits in correct order <u>Examiner's Comments</u> No hint was given about whether or not to use the work done for Q6(a), however it was disappointing to see so many candidates embark on substitution methods, ignoring (or not seeing) the connection between the two parts.

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	6		
12			DR $\int_{-1}^{11} \frac{1}{\sqrt{x^2+6x+13}}\,dx$ $x^2+6x+13 \equiv (x+3)^2+4$ $\Rightarrow \int_{-1}^{11} \frac{1}{\sqrt{x^2+6x+13}}\,dx = \int_{-1}^{11} \frac{1}{\sqrt{(x+3)^2+4}}\,dx$ $= \left[\ln\left((x+3)+\sqrt{(x+3)^2+4}\right)\right]_{-1}^{11}$ $= \ln\left(14+\sqrt{200}\right) - \ln\left(2+\sqrt{8}\right)$ $= \ln\left(\frac{7+5\sqrt{2}}{1+\sqrt{2}}\right) = \ln\left(3+2\sqrt{2}\right)$ i.e. $p = 3$, $q = 2$	M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 1.1) A1 (AO 1.1) [7]	Attempt to complete the square In a suitable form to integrate Use standard form Use correct limits to obtain answer in lns	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		a	$f(x)=\ln(2+x) \Rightarrow f'(x)=\frac{1}{2+x} \Rightarrow f'(0)=\frac{1}{2}$	M1 (AO 1.1) A1 (AO 1.1) [2]	Differentiation	
		b	$f'(x)=\frac{1}{2+x} \Rightarrow f''(x)=-\frac{1}{(2+x)^2}$ $\Rightarrow f''(0)=\frac{-1}{(2+0)^2}=-\frac{1}{4}$	M1 (AO 1.1) A1 (AO 2.1) [2]	Differentiation AG	
		c	f(0) = ln 2 $f(x)=f(0)+xf'(0)+\frac{x^2}{2}f''(0)$ $=\ln 2+\frac{1}{2}x-\frac{1}{8}x^2$	B1 (AO 1.1) B1 (AO 1.1) B1FT (AO 1.1) [3]	soi 2 two terms correct 3rd term also correct. FT their values	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
14		a	$\cosh(\mathrm{i}z) = \dfrac{\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z}}{2}$ $= \dfrac{\cos z + \mathrm{i}\sin z + \cos z - \mathrm{i}\sin z}{2}$ $= \dfrac{2\cos z}{2} = \cos z$ **AG**	M1 (AO 2.1) M1 (AO 1.1) A1 (AO 2.1) [3]	Use of correct exponential form for cosh Correct use of Euler's formula at least once Both M marks must be awarded. Must have cosh(iz) = or LHS =	Proof must be complete for A1	
		b	$\cos z = 2 \Rightarrow \cosh(\mathrm{i}z) = 2 \Rightarrow z = (\cosh^{-1}2)/\mathrm{i}$ $= -\mathrm{i}\ln(2+\sqrt{3})$	M1 (AO 3.1a) A1 (AO 1.1) [2]	±inside or outside the ln (ie allow eg iln(2 + $\sqrt{3}$) or iln(2 – $\sqrt{3}$) and condone eg ±iln(2 + $\sqrt{3}$) www)	or $2\pi n$ ± iln(2 + $\sqrt{3}$) for any integer n	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
15			$\cosh^2 x - \sinh^2 x = 1$	M1 (AO 1.1a)	Use of identity to leave an equation in either just coshx or just sinhx	$4\cosh^4 x - 45\cosh^2 x + 50 = 0$	
			$2\sinh^2 x + 5\sinh x - 3 = 0$	M1 (AO 1.1)	Reduction to 3 term quadratic in sinhx or cosh2x	$\cosh^2 x = 5/4$ or 10	
			$\sinh x = {}^1/_2$ or -3	A1 (AO 1.1)			
			$x = \ln\left(\frac{1}{2} + \sqrt{\frac{5}{4}}\right)$	M1 (AO 1.1)	Use of ln formula for $\sinh^{-1}$ or $\cosh^{-1}$	Do not allow eg $\cosh^{-1}(-\sqrt{10})$ unless this is rejected (NB eg ln $(3 - \sqrt{10})$ is not real). If using $\cosh^{-1}$ "rogue" solutions must be convincingly rejected. Most likely to see ln $(3 + \sqrt{10})$	
			$x = \ln\left(\frac{1}{2} + \frac{1}{2}\sqrt{5}\right)$	A1 (AO 1.1)	Must be in the correct form but allow $\ln\left(\frac{1+\sqrt{5}}{2}\right)$		
			$x = \ln\left(-3 + \sqrt{10}\right)$	A1 (AO 1.1) [6]	$p = ½, q = ½, r = 5$ $p = -3, q = 1, r = 10$		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		a	$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots$ $\Rightarrow \ln(1+3x^2)=(3x^2)-\frac{(3x^2)^2}{2}+\frac{(3x^2)^3}{3}-\frac{(3x^2)^4}{4}+\ldots$ $=3x^2-\frac{9x^4}{2}+9x^6-\frac{81x^8}{4}+\ldots$	M1 (AO1.1a) A1 (AO1.1) [2]	Substitute $3x^2$ into standard formula or $\frac{27x^6}{3}$	
		b	Valid for $3x^2 \leqslant 1 \Rightarrow \lvert x\rvert \leqslant \frac{1}{\sqrt{3}}$ oe	B1 (AO 2.2a) [1]		

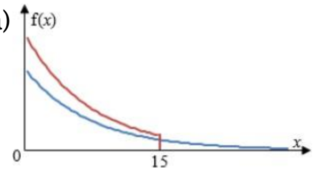
Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	Substitute $x = \frac{1}{2}$ $\Rightarrow \ln(1+3x^2) = \ln\left(1+\frac{3}{4}\right)$ $= 3\left(\frac{1}{2}\right)^2 - \frac{9}{2}\left(\frac{1}{2}\right)^4 + \frac{27}{3}\left(\frac{1}{2}\right)^6 - \frac{81}{4}\left(\frac{1}{2}\right)^8 + \ldots.$ $\Rightarrow \ln\left(\frac{7}{4}\right) = 3\left(\frac{1}{2}\right)^2\left(1 - \frac{3}{2}\left(\frac{1}{2}\right)^2 + \frac{3^2}{3}\left(\frac{1}{2}\right)^4 - \frac{3^3}{4}\left(\frac{1}{2}\right)^6 + \ldots\right)$ $\Rightarrow \frac{3}{2}\left(\frac{1}{2}\right)^2 + \frac{9}{3}\left(\frac{1}{2}\right)^4 - \frac{3^3}{4}\left(\frac{1}{2}\right)^6 + \ldots = 1 - \frac{4}{3}\ln\left(\frac{7}{4}\right)$	M1 (AO3.1a) M1 (AO1.1) A1 (AO3.2a) [3]	Deducing the required value for x Rearranging either series (with either ½ or x) to show the relationship with the other	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance
17			DR		
			$\frac{x+1}{x^3-x^2+x-1}=\frac{x+1}{(x^2+1)(x-1)}$	M1 (AO3.1a)	Factorise denominator
			$=\frac{Ax+B}{(x^2+1)}+\frac{C}{(x-1)}$	M1 (AO2.1)	Split into partial fractions
			$\Rightarrow (Ax+B)(x-1)+C(x^2+1)=x+1$ $x=1$: $2C=2 \Rightarrow C=1$ $x=0$: $-B+C=1 \Rightarrow B=0$ $x=2$: $2A+5=3 \Rightarrow A=1$	M1 (AO3.1a)	Valid method for finding A, B and C
			$\Rightarrow \frac{x+1}{(x^2+1)(x-1)}=\frac{1}{(x-1)}-\frac{x}{(x^2+1)}$	A1 (AO1.1)	
			$\Rightarrow \int_2^3 \frac{x+1}{x^3-x^2+x-1}\,dx=\int_2^3\left(\frac{1}{(x-1)}-\frac{x}{(x^2+1)}\right)dx$ $\left[\ln(x-1)-\frac{1}{2}\ln(x^2+1)\right]_2^3=\ln 2-\frac{1}{2}\ln 10+\frac{1}{2}\ln 5$	M1 (AO1.1)	Substituting correct limits into their integral of the form $\alpha\ln(x-1)+\beta\ln(x^2+1)$
			$=\frac{1}{2}\ln 2$	A1 (AO2.2a) [6]	oe in correct form (not eg $\ln\sqrt{2}$)
			Total	6	

Question			Answer/Indicative content	Marks	Part marks and guidance	
18		a	DR $x^2 - 2x + 10 = (x - 1)^2 + 9$ $\Rightarrow = \int_1^4 \frac{1}{(x-1)^2+9} dx = \frac{1}{3}\left[\tan^{-1}\left(\frac{x-1}{3}\right)\right]_1^4$ $= \frac{1}{3}\left(\frac{\pi}{4} - 0\right) = \frac{\pi}{12}$	M1 (AO3.1a) A1 (AO1.1a) A1 (AO1.1) [3]	Completing the square Ignore limits	
		b	DR $\text{Mean value} = \frac{1}{\left(\frac{1}{2} - 0\right)} \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} dx$ $= 2\left[\sin^{-1} x\right]_0^{\frac{1}{2}} = 2\left(\frac{1}{6}\pi - 0\right)$ $= \frac{1}{3}\pi$	M1 (AO1.1a) M1 (AO1.1) A1 (AO1.1) [3]	Uses formula for mean value Uses formula for integral and attempts substitution of limits	
			Total	6		

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance		
19		a	DR $\int_0^\infty x\lambda e^{-\lambda x}dx$ $=\left[-xe^{-\lambda x}\right]_0^\infty-\int_0^\infty -e^{-\lambda x}dx$ $=\lim_{k\to\infty}\left(\left[-xe^{-\lambda x}-\frac{1}{\lambda}e^{-\lambda x}\right]_0^k\right)$ $=\frac{1}{\lambda}$ so $\lambda=\frac{1}{5.0}=0.2$	M1 (AO 1.1a) M1 (AO 1.1) E1 (AO 2.4) A1 (AO 1.1) [4]	$\int_{20}^\infty 0.2e^{-0.2x}dx$ Use Parts used explicitly, $e^{-\lambda}$ integrated or equivalent consideration of limits Correctly obtain $\frac{1}{\lambda}$ and λ = 0.2, www	accept $=0-\left[\frac{1}{0.2}e^{-0.2x}\right]_0^\infty$	
		b	$\int_{20}^\infty 0.2e^{-0.2x}dx$ = 0.0183156...	M1 (AO 3.1b) A1 (AO 3.4) [2]	Or $1-\int_0^{20} 0.2e^{-0.2x}dx$ BC Awrt 0.0183		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	(a)  (b)	B1 (AO 1.1) [1] M1 (AO 3.5c) A1 (AO 3.3) [2]	Correct shape for $y = \lambda e^{-\lambda x}$, asymptotic to x−axis only Any curve truncated at 15 More than half above their curve from part (i)	Don't need vertical line i.e., awareness that total area is the same	
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		a	$V=\pi\int_{\sqrt{p}}^{\sqrt{3p}}\left(\frac{1}{\sqrt{p+x^2}}\right)^2\mathrm{d}x$ soi $\int\left(\frac{1}{\sqrt{p+x^2}}\right)^2\mathrm{d}x=\int\frac{1}{\left(\sqrt{p}\right)^2+x^2}\mathrm{d}x$ $=\frac{1}{\sqrt{p}}\tan^{-1}\left(\frac{x}{\sqrt{p}}\right)$ $\left[\tan^{-1}\left(\frac{x}{\sqrt{p}}\right)\right]_{\sqrt{p}}^{\sqrt{3p}}=\tan^{-1}\left(\frac{\sqrt{3p}}{\sqrt{p}}\right)-\tan^{-1}\left(\frac{\sqrt{p}}{\sqrt{p}}\right)$ $V=\pi\times\frac{1}{\sqrt{p}}\left(\frac{\pi}{3}-\frac{\pi}{4}\right)=\frac{\pi^2}{12\sqrt{p}}$ oe	B1 (AO 1.2) M1 (AO 2.2a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [5]	Correct limits and function substituted in. Condone missing π. Writing in integrable form. If not seen condone incorrect constant outside or use of p for $\sqrt{p}$ inside for M1 Correct use of limits in an integrated expression of the form $a\tan^{-1}(bx)$	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$p = 1$,	B1 (AO 2.2a)		At least one of V_{max} and V_{min} must be clearly identified as such	
			$V \propto \frac{1}{\sqrt{p}} \Rightarrow V_{max} = \frac{\pi^2}{12}$	M1 (AO 2.2a)	Either seen		
				A1 (AO 1.1)			
			$\sqrt{3p} = \sqrt{48} \Rightarrow p = 16$	[3]		If neither identified then B0M1A1 or B1M1A0 are available.	
			$p = 16 \Rightarrow V_{min} = \frac{\pi^2}{48}$				
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		a	$\int 2\tan x\,dx = -2\ln(\cos x)$ or $2\ln(\sec x)$	B1 (AO 1.1a)		Limits can be ignored for M1A1
			$-2\left[\ln(\cos x)\right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} = -2\left(\ln\cos\frac{\pi}{3} - \ln\cos\frac{\pi}{4}\right)$	M1 (AO 1.1)	Correct use of limits in integral of the form $a\ln f(x)$ where $f(x)$ is trigonometric	
			$= -2\left(\ln\frac{1}{2} - \ln\frac{1}{\sqrt{2}}\right) = -2\ln 2^{-\frac{1}{2}} = \ln 2$	A1 (AO 1.1)		
				[3]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	$\int_0^z 2\tan x\,dx = -2\ln\cos z$ So $\int_0^{\frac{\pi}{2}} 2\tan x\,dx = \lim_{z\to\frac{\pi}{2}}(-2\ln\cos z)$ but $z \to \frac{\pi}{2} \Rightarrow \cos z \to 0 \Rightarrow \ln\cos z \to -\infty$ so the integral is undefined.	M1 (AO 2.1) A1 (AO 2.2a) [2]	Using a variable to denote the upper limit and finding the integral. Could be their incorrect integral from (a) or ln0 is undefined but argument must be complete	Or $\int_0^z 2\tan x\,dx = 2\ln\sec z$ Or $\int_0^{\frac{\pi}{2}} 2\tan x\,dx = \lim_{z\to\frac{\pi}{2}}(2\ln\sec z)$ but $z \to \frac{\pi}{2} \Rightarrow \sec z \to \infty$ $\Rightarrow \ln\sec z \to \infty$ so the integral is undefined.
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		a	(i) $\sec\theta = \frac{r}{x}$	B1 (AO1.1)		
			$\Rightarrow r = 2 - \frac{r}{x}$	B1 (AO3.1a)		
			$\Rightarrow r = \frac{2x}{x+1}$	B1 (AO3.1a)		
			$y^2 = r^2 - x^2 \Rightarrow y^2 = \left(\frac{2x}{x+1}\right)^2 - x^2$	A1 (AO2.1)	AG	
				[4]		
			(ii) Because the curve is y^2 =...... and so $x = p$ gives $y = \pm q$	B1 (AO2.4)		
				[1]		
			(iii) $a = -1$	B1 (AO2.2a)	Accept $x = -1$	
				[1]		
		b	$V = \pi \int y^2 \, dx$	M1 (AO1.1a)	Use of formula	
			$= \pi\int\left(\left(\frac{2x}{x+1}\right)^2 - x^2\right)dx = \pi\int\left(4 - x^2 - 4\left(\frac{2x+2}{x^2+2x+1}\right) + \frac{4}{(x+1)^2}\right)dx$	M1 (AO3.1a)	any equivalent form that can be integrated	
				A1 (AO1.1)		
			$\pi\left[4x - \frac{x^3}{3} - 4\ln(x+1)^2 - \frac{4}{(x+1)}\right]$	M1 (AO3.1a)	Attempt to integrate	
				A1 (AO1.1)		
			Limits are 0 and 1	B1 (AO3.1a)		
			$\Rightarrow V = \pi\left(\left(4 - \frac{1}{3} - 4\ln 4 - 2\right) - (0 - 0 - 0 - 4)\right)$	M1 (AO1.1)		
			$= \pi\left(\frac{17}{3} - 4\ln 4\right)$	A1 (AO1.1)	4ln4 and 8ln2 are equivalent	
				[8]		
			Total	14		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23			$\frac{dQ}{dt} = -k(1+Q^2) \Rightarrow \int \frac{dQ}{1+Q^2} = -k\int dt$	B1 (AO3.3) M1 (AO3.1b)		
			$\Rightarrow \tan^{-1} Q = c - kt$	A1 (AO1.1)		
			$t = 0, Q = 100 \Rightarrow c = \tan^{-1} 100$	A1 (AO3.4)		
			$t = 100,\ Q = 50$ $\Rightarrow \tan^{-1} 50 = \tan^{-1} 100 - 100k$	A1 (AO3.4)		
			$\Rightarrow k = \frac{1}{100}\left(\tan^{-1}100 - \tan^{-1}50\right) \approx 0.0001$	M1 (AO1.1a)		
			When $t = 400,\ Q = 20$	A1 (AO3.4)		
			20 g	A1 (AO3.2b)	Must have unit	
				[8]		
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		a	$f(x)=(1+a\sin x)e^{bx}=\left(1+a\left(x-\frac{x^3}{6}\right)\right)\left(1+bx+\frac{b^2x^2}{2}\right)$ $=1+bx+\frac{b^2x^2}{2}+ax+abx^2=1+(a+b)x+x^2\left(\frac{b^2}{2}+ab\right)$ $\approx 1+2x+\frac{3}{2}x^2$ $\Rightarrow a+b=2,\ \frac{b^2}{2}+ab=\frac{3}{2}$ $\Rightarrow b^2+2b(2-b)=3 \Rightarrow b^2-4b+3=0$ $\Rightarrow b=1, 3 \Rightarrow a=1, -1$ $\Rightarrow (-1, 3)$	B1 (AO1.2) B1 (AO1.2) M1 (AO1.1a) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [6]	Sin expansion Exponential expansion Multiply out and compare coefficients Both equations Solve As $a<0$	
		b	There is no restriction on x because there is no restriction on the maclaurin's expansion for $\sin x$ and e^x.	B1 (AO2.3) [1]		
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
25			$I=\int_{3/2}^{5/2}\frac{1}{\sqrt{4x^2-12x+13}}\mathrm{d}x=\int_{3/2}^{5/2}\frac{1}{\sqrt{(2x-3)^2+4}}\mathrm{d}x$ Substitute $2x - 3 = 2\sinh\theta$ $\Rightarrow dx = \cosh\theta d\theta$, $(2x-3)^2 + 4 = 4\cosh^2\theta$ When $x=\frac{3}{2}$, $\theta = 0$. When $x=\frac{5}{2}$, $\sinh\theta = 1$ $\Rightarrow I=\int_{0}^{\sinh^{-1}1}\frac{1}{2\cosh\theta}\cosh\theta\mathrm{d}\theta=\frac{1}{2}[\theta]_0^{\sinh^{-1}1}$ $=\frac{1}{2}\sinh^{-1}1=\frac{1}{2}\ln\left(1+\sqrt{2}\right)$ OR $\int_{3/2}^{5/2}\frac{1}{\sqrt{4(x-\frac{3}{2})^2-9+13}}\mathrm{d}x=\frac{1}{2}\int_{3/2}^{5/2}\frac{1}{\sqrt{(x-\frac{3}{2})^2+1}}\mathrm{d}x=$ $\frac{1}{2}\int_0^1\frac{1}{\sqrt{u^2+1}}\mathrm{d}x=\frac{1}{2}\left[\ln(u+\sqrt{u^2+1}\right]=\frac{1}{2}\ln(1+\sqrt{2})$	M1 (AO2.1) M1 (AO3.1a) A1 (AO1.1) A1 (AO2.2a) M1 (AO1.1) A1 (AO1.1) [6]	Completing the square Substitution Correct reduction (ignore limits) Correct integral (ignore limits) Limits $u=x-\frac{3}{2}$ du = dx $x=\frac{5}{2}$, $u=1$ $x=\frac{3}{2}$, $u=0$	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance		
26			DR $f(x) = x^3 - 3x^2 + 4x - 12$ and $f(3) = 0$	M1 (AO 3.1a)	Attempt to find a factor of the denominator	Could be by factorising pairs of terms	
			$x - 3$ is a factor	A1 (AO 2.2a)			
			$f(x) = (x-3)(x^2 + 4)$	A1 (AO 2.2a)	Deduce by inspection but may see symbolic division oe.		
			$\frac{2x^2+3x-1}{x^3-3x^2+4x-12} = \frac{A}{x-3} + \frac{Bx+C}{x^2+4}$	B1 (AO 3.1a)	Correct partial fraction expression using their factorised cubic		
			$2x^2 + 3x - 1 = A(x^2 + 4) + (Bx + C)(x - 3)$	M1 (AO 3.1a)	A correct method for finding A, B or C (no terms omitted)		
			$A = 2$	A1 (AO 1.1)			
			$C = 3$	A1 (AO 1.1)			
			$B = 0$	A1 (AO 1.1)			
			$\int \frac{1}{x-3} dx = \ln\|x-3\|$	B1 (AO 1.1a)			
			$\int \frac{1}{x^2+4} dx = \frac{1}{2}\tan^{-1}\frac{x}{2}$	B1 (AO 1.1)	Ignore multiplicative or additive constant. Condone omitted modulus signs.		
			$2\ln\|2-3\| + \frac{3}{2}\tan^{-1}\frac{2}{2} - \left(2\ln\|0-3\| + \frac{3}{2}\tan^{-1}\frac{0}{2}\right)$	M1 (AO 1.1)	Allow for (any one of) their linear factor(s)		
			$-2\ln 3 + \frac{3}{2} \times \frac{\pi}{4} = \frac{3}{8}\pi - \ln 9$	A1 (AO 1.1) [12]	Ignore		

Question			Answer/Indicative content	Marks	Part marks and guidance		
					additive constant. Allow for their (numerical) *a*. Correct substitution of limits (and subtraction) into an integral containing ln and $\tan^{-1}$. Condone $\tan^{-1}0$ missing. AG Must see evidence of use of laws of logs.		
			Total	12			

Question			Answer/Indicative content	Marks	Part marks and guidance		
27			DR $\dfrac{1}{(1.5-0)}\displaystyle\int_0^{1.5}\frac{1}{\sqrt{9-t^2}}\mathrm{d}t$ $\left[\sin^{-1}\dfrac{t}{3}\right]$ $\sin^{-1}\dfrac{1.5}{3}$ $\dfrac{\pi}{6}$ So average velocity is 0.35 cms^{-1} along *OA*	B1 (AO 3.4) M1 (AO 3.4) A1 (AO 3.4) A1 (AO 1.1) A1 (AO 3.4) [5]	Using the formula for mean value Ignore limits Correct use of limits. Soi by answer correct unit must be present or accept awrt 0.349 o $\dfrac{\pi}{9}$ r	May be implied by correct integral and later division by 1.5 o $\left[\cos^{-1}\dfrac{t}{3}\right]$ r o $\cos^{-1}\dfrac{1.5}{3}-\cos^{-1}0$ r	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	$y=\tan^{-1}\left(\frac{x}{x+1}\right)\Rightarrow\frac{dy}{dx}=\frac{1}{1+\left(\frac{x}{x+1}\right)^2}\cdot\frac{(x+1).1-x.1}{(x+1)^2}$	M1	Attempt to apply chain rule and deal with quotient	$\tan y=\frac{x}{x+1}$ $\Rightarrow\sec^2 y\frac{dy}{dx}=\frac{1}{(x+1)^2}$ M1 for implicit diffn
			$=\frac{1}{(x+1)^2+x^2}=\frac{1}{2x^2+2x+1}$	A1	Soi	A1 for expression for y' $\Rightarrow\sec^2 y\frac{d^2y}{dx^2}+2\sec^2 y\tan y\left(\frac{dy}{dx}\right)^2=-\frac{2}{(x+1)^3}$ A1
			$\Rightarrow\frac{d^2y}{dx^2}=-\frac{4x+2}{\left(2x^2+2x+1\right)^2}$	A1		Substitute $x=0$, $\tan y=0$, $\sec y=1$
			$(\text{When } x=0,)\ \frac{d^2y}{dx^2}=\left(-\frac{2}{(1)^2}\right)=-2$	A1	www AG	$\Rightarrow\frac{d^2y}{dx^2}=-2$ A1
				[4]	Examiner's Comments Some candidates found part (i) very difficult and several used additional sheets, or even additional answer books, for this part. The candidates who realised that they had to use the chain rule were usually fine, apart from occasional sign errors, but many candidates ignored the change of variable and just wrote the answer as $\frac{1}{1+\left(\frac{x}{1+x}\right)^2}$, often spending time trying to simplify this and then using the quotient rule to try to find the second	

Question			Answer/Indicative content	Marks	Part marks and guidance		
					derivative.		
		ii	$y_0 = 0$, $y_0' = 1$ www in part (i) soi $\Rightarrow$ (y =) $x - x^2$	DB1 B1ft [2]	Use values from (i)	Dependent on correct working in (i) i.e. $y = 0 + kx - x^2$ where k is *their* y'	
					Examiner's Comments For part (ii) some candidates tried replacing x by $\left(\frac{x}{1+x}\right)$ in the Maclaurin expansion of $\tan^{-1}x$, but most realised that having found the first and second derivatives they knew enough information to write down the expansion as far as the term in x^2. An alternative approach was to rewrite the equation as and to differentiate implicitly. Some immediately found $\frac{dy}{dx}$ when $x = 0$, then moved on to find $\frac{d^2y}{dx^2}$ implicitly and hence the value of $\frac{d^2y}{dx^2}$ when $x = 0$. Others substituted for $\sec^2 y$ back into x first and had to do some heavy algebraic manipulation to achieve the given answer; inevitably many made errors in this working.		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
29			$I=\int_3^4 \frac{1}{x^2-6x+10}\,\mathrm{d}x=\int_3^4 \frac{1}{(x-3)^2+1}\,\mathrm{d}x$	M1 A1	Complete the square. Denominat or	Sight of $(x \pm 3)^2 + 1$
			From formula book: $I=\left[\tan^{-1}(x-3)\right]_3^4=\tan^{-1}1 \pm \tan^{-1}0$	M1	Use standard form and apply limits	Accept $\tan^{-1}(x \pm 3)$
			$=\frac{\pi}{4}$	A1	www	SC correct answer only B1
				[4]	Examiner's Comments Most completed the square correctly but some could not then use this to carry out the integration. Several sign errors, some who could not decide between $\tan(x-3)$ and $\tan^{-1}(x-3)$ and some who thought that the answer needed to involve $2\ln(x^2\ 6x+10)$. A small minority who managed the integration gave an answer in degrees rather than radians.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
30		a	$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sinh^{-1}(2x)\right)$ $=\frac{1}{\sqrt{x^2+(\frac{1}{2})^2}}$ $=\frac{1}{\sqrt{x^2+(\frac{1}{2})^2}}\times\frac{2}{2}=\frac{2}{\sqrt{4x^2+1}}$	M1(AO 1.1a) E1(2.1) [2]	Use formulae from formulae booklet AG At least one intermediate step must be seen	OR M1 $\sinh y = 2x$ so $\frac{\mathrm{d}y}{\mathrm{d}x}\cosh y = 2$ **E1** $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{2}{\cosh y}=\frac{2}{\sqrt{4x^2+1}}$	
		b	$2-2x+x^2=1+(x-1)^2$ $\Rightarrow\int\frac{1}{\sqrt{1+(x-1)^2}}\,\mathrm{d}x=\left[\sinh^{-1}(x-1)\right]+c$	B1(AO 3.1a) M1(AO 1.1) A1(AO1.1) [3]	Complete square soi Use correct form oe		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
31		a	$\sin 2x = 2x - \frac{4}{3}x^3 \ldots$ and $\sin 4x = 4x - \frac{32}{3}x^3 \ldots$ $\left(x - \frac{1}{6}x^3\right)\left(2x - \frac{4}{3}x^3\right)\left(4x - \frac{32}{3}x^3\right)$ $= 8x^3 \square - 28x^5$	M1(AO2.1) A1(AO1.1a) M1(AO1.1) A1(AO1.1) [4]	$x \rightarrow 2x$ and $x \rightarrow 4x$ in $\sin x$ expansion Attempt complete expansion These terms and no others	
		b	$(x)(56x^4 - 16x^2 + 1) = 0$ oe $x^2 = \dfrac{16 \pm \sqrt{32}}{112}$ $x = \sqrt{\dfrac{4 - \sqrt{2}}{28}}$ or equivalent exact surd.	M1(AO1.1) M1(AO3.1a) A1(AO1.1) [3]	Substitute and rearrange Solve as quadratic in x^2 Select correct answer	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
32			DR $-2xe^{-x} + \int 2e^{-x}\,dx$ Evaluation of their F[x] using both limits [0 – 0] – [0 – 2] 2	*M1(AO1.1) A1(AO1.1) dep*M1(AO1.1) A1(AO1.1) [4]	Allow sign errors only Must be seen F[x] = $-2xe^{-x}$□$-2e^{-x}$	e.g. M1 for $\pm 2xe^{-x} \pm \int 2e^{-x}\,dx$ Evaluation must be seen	
			Total	4			
33			**DR** $[V=]\pi\displaystyle\int_0^4\left(\frac{8}{\sqrt{16+x^2}}\right)^2 dx$ $64\pi\times\frac{1}{4}\tan^{-1}\left(\frac{x}{4}\right)$ $16\pi\tan^{-1}(1) - 16\pi\tan^{-1}(0)$ $= 4\pi^2$	B1(AO1.1) M1(AO1.2) M1(AO1.1) A1(AO1.1) [4]		Must be seen Using formulae booklet Substitution of correct limits must be seen	
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance	
34			$f(x)=\frac{x(x-1)}{(x+1)(x^2+1)}\equiv\frac{A}{(x+1)}+\frac{Bx+C}{(x^2+1)}$ $\Rightarrow A(x^2+1)+(Bx+C)(x+1)\equiv x(x-1)$	M1	Correct partial fractions	
			For e.g. equate coefficients $\Rightarrow A+B=1,\ B+C=-1,\ A+C=0$	M1	Dep on 1st M	Or sub values of x or division
			$\Rightarrow A=1,\ B=0,\ C=-1$ $\Rightarrow (f(x))=\frac{1}{(x+1)}-\frac{1}{(x^2+1)}$	A1	Dep on both M marks.	
			$\Rightarrow\int_0^1 f(x)dx=\int_0^1\left(\frac{1}{(x+1)}-\frac{1}{(x^2+1)}\right)dx$	B1	ft for integrating 1st term correctly $(A/(x+1)\ A\neq0)$	
			$=\left[\ln(1+x)-\tan^{-1}x\right]_0^1=\ln 2-\frac{\pi}{4}$	B1	ft for subsequent term(s) correctly	
				B1	in exact form as dep on both previous B marks	Examiner's Comments The partial fractions part was nearly always attempted in the correct manner with almost all of these then going on to the correct result. Many candidates did not explicitly state the partial fractions but this was condoned if they were then seen in the integration. A surprisingly high number of candidates did not spot the arc tan integration and gave some version of ln instead. Given that this is a FP2 paper they should expect to be integrating functions that are uniquely in this specification.
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	$\frac{dy}{dx} = \frac{2}{1+4x^2}$	B1	For first diffn	
		i	$\frac{d^2y}{dx^2} = -2\left(1+4x^2\right)^{-2} \times 8x$ $= \frac{-16x}{\left(1+4x^2\right)^2} = -4x\left(\frac{dy}{dx}\right)^2$	M1	Diffn again and making comparison	Examiner's Comments There were a number of methods employed to differentiate $\tan^{-1} 2x$: • use the standard result from the formula sheet plus chain rule, • rearrange to $\tan y = 2x$ and differentiate implicitly, • rearrange to $x = \frac{1}{2\tan y}$ and find $\frac{1}{\frac{dx}{dy}}$. Almost all candidates obtained the required answer although the numerator of 2 was missing from those who found problems using the chain rule. The second differential posed few problems for most and then followed the algebraic proof to the required answer. Candidates should be aware that proofs of this type must contain no algebraic errors and it was a little surprising that a significant minority made careless slips; in almost all of these cases the algebraic slip was then mirrored in the compared expression.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	When $x=0,\ y=0, \frac{dy}{dx}=2, \frac{d^2y}{dx^2}=0$	B1	Soi by final answer	See below for alternative differentiation.
		ii	$\Rightarrow \frac{d^3y}{dx^3}+4\left(\frac{dy}{dx}\right)^2+8x\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2}=0$	M1	Differentiate the equation given	
		ii	When $x=0,\ \frac{d^3y}{dx^3}+16=0$	A1	For – 16 www	SC4 Final formula from formula book including sight of $(2x)^3$
		ii	$\Rightarrow (y)=2x-\frac{8x^3}{3}$	A1	Final answer	SC2 if final form only seen (i.e. no working) SC0 if final form only and wrong Examiner's Comments The expression required in part (i) was intended as a hint for candidates as to the most efficient form for finding the third differential; most ignored the suggestion and used the quotient rule to attempt the differentiation. Unsurprisingly, there were a considerable number of slips when handling this messy expression and it must be appreciated that errors at this stage will be penalised in the later part of this solution. In the final Maclaurin expansion it is expected that all coefficients are presented in their simplest form and answers such as $\frac{-16}{3!}x^3$ are not acceptable.
		iii	$x=\frac{1}{2}\Rightarrow \tan^{-1}1=\frac{\pi}{4}$	B1	soi	
		iii	In series $x=1-\frac{1}{3}=\frac{2}{3}$			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow$ Estimate for $\pi = \frac{8}{3} = 2.666....$	B1	For showing to 1sf $\pi = 3$ which is correct but to 2sf $\pi = 2.7$ which is not. www	
		iii	which, correct to 1sf, = 3 Alternative differentiation: $y'' = \frac{-16x}{(1+4x^2)^2} \Rightarrow$ $y''' = \frac{-16(1+4x^2)^2 + 256(1+4x^2)x^2}{(1+4x^2)^4}$ or $y'' = -16x(1+4x^2) \Rightarrow y''' = 256x^2(1+4x^2)^{-3} - 16(1+4x^2)^{-2}$		M1 is for two terms on top M1 is for 2 terms	Examiner's Comments This short question on using the series appropriately was not well done. Most realised that $\tan^{-1}\left(2\times\frac{1}{2}\right) = \frac{\pi}{4}$ was required but ignored the word only which implied that not only should the values be identical when rounded to one significant figure but that they be different when rounded to two significant figures. Only the very best candidates were able to appreciate that this extra element was required.
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
36			$\ln(1+y)=y-\frac{y^2}{2}+\frac{y^3}{3}-\ldots$	B1	Soi. Allow an expansion in x	
			$\sin x=x-\frac{x^3}{6}+\ldots$	B1	Soi	
			$\ln(1+\sin x)=\left(x-\frac{x^3}{6}\right)-\frac{1}{2}\left(x-\frac{x^3}{6}\right)^2+\frac{1}{3}\left(x-\frac{x^3}{6}\right)^3-\ldots$ $=x-\frac{1}{2}x^2+x^3\left(\frac{1}{3}-\frac{1}{6}\right)$	M1	For combining series, even if wrong. Must include at least the cubic bracket.	
			$=x-\frac{1}{2}x^2+\frac{1}{6}x^3$	A1	Ignore further terms www accept 3! for 6	
			Alternative using Maclaurin general formula			
			$f(x)=\ln(1+\sin x)$ $\quad f(0)=0$ $f'(x)=\frac{\cos x}{(1+\sin x)}$ $\quad f'(0)=1$	B1	For f '(x)	
			$f''(x)=\frac{-1}{(1+\sin x)}$ $\quad f''(0)=-1$ $f'''(x)=\frac{\cos x}{(1+\sin x)^2}$ $\quad f'''(0)=1$	B1	For (not necessarily simplified) f ''(x) and f ''(0) www	
			Maclaurin: $f(x)=f(0)+f'(0)x+\frac{f''(0)x^2}{2}+\frac{f'''(0)x^3}{6}$	M1	For correct formula up to 4th term and substituting *their* values	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\Rightarrow f(x) = x - \frac{1}{2}x^2 + \frac{1}{6}x^3$	A1	Accept 3! for 6 Examiner's Comments In this question it was indicated that candidates should use the standard expansions for $\ln(1+x)$ and $\sin x$. Those that did so found the question straightforward. The only recurrent problem was the failure to take the ln expansion to the cubic term. In this question there was an alternative method, which was to find the 1st, 2nd and 3rd derivatives and substitute into the general Maclaurin series. This was perfectly acceptable and many candidates obtained full marks. However, most found this method rather longer and the process of differentiating increasingly complicated expressions resulted in most candidates making an error and therefore not achieving the correct result. It was not helped by the fact that many did not simplify the 2nd derivative before differentiation again.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
37			$\int_{\frac{1}{2}}^{1} \frac{1}{\sqrt{2x-x^2}}\,dx = \int_{\frac{1}{2}}^{1} \frac{1}{\sqrt{1+2x-x^2-1}}\,dx$	M1	Completing the square on given function	
			$= \int_{\frac{1}{2}}^{1} \frac{1}{\sqrt{1-(1-x)^2}}\,dx$	A1		Or $= \int_{\frac{1}{2}}^{1} \frac{1}{\sqrt{1-(x-1)^2}}\,dx$ $= \left[\sin^{-1}(x-1)\right]_{\frac{1}{2}}^{1} = \left(0 - -\frac{\pi}{6}\right) = \frac{\pi}{6}$
			$= \left[-\sin^{-1}(1-x)\right]_{\frac{1}{2}}^{1}$	M1	By substitution or using standard form where completed square is of form $1-(1\pm x)^2$	
				A1	Correct result of integration. Ignore limits	
			$= -\left(0 - \frac{\pi}{6}\right) = \frac{\pi}{6}$	A1	Examiner's Comments Unfortunately the answers to this part were marred by the inability to complete the square, a procedure that is first tested in GCSE. The other common error that caused a loss of marks was to fail to deal effectively with a double negative which was often just ignored on the assumption that the answer was positive. A very neat alternative method to this question was to make the substitution $x = 1 + \sin\theta$. This was seen rarely but always produced the correct answer and full marks.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
38		i	$\frac{\cos x}{1+\sin x} - \frac{-\sin x}{\cos x}$ or $\frac{\cos x}{1+\sin x} + \frac{\sin x}{\cos x}$ (I)	B2	Each half (including 'middle' sign) scores B1	
		i	$\frac{+/-\cos^2 x + /-\sin x(1+\sin x)}{(1+\sin x)\cos x}$	M1	Combine, provided derivative was of form $f'(x)/f(x)$	Allow only variations num signs
		i	$\frac{1+\sin x}{\cos x(1+\sin x)} = \frac{1}{\cos x}$ www **AG**	A1	$\cos^2 x + \sin^2 x = 1$ in intermediate step required	
		i	(II) Change to $\ln\left(\frac{1+\sin x}{\cos x}\right)$	B1		
		i	Change to ln(sec x + tan x)	B1	Not $\ln(\frac{1}{\cos x} + \tan x)$ Examiner's Comments Three different methods were seen for this. The normal one, using the straightforward chain rule, proved successful for the majority. A few completed an initial double change into ln(sec x + tan x) and, again, were usually successful with the differentiation. An even smaller number changed the expression initially to just ln $\left(\frac{1+\sin x}{\cos x}\right)$ but then had to cope with the differentiation of a quotient, which often proved a rather overwhelming prospect.	
		i	Diff as $\frac{\text{attempt at } \frac{d}{dx}(\sec x + \tan x}{\sec x + \tan x}$	M1		
		i	Reduce to $\sec x = \frac{1}{\cos x}$	A1		
		i	(III) Change to $\ln\left(\frac{1+\sin x}{\cos x}\right)$	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	Diff as $\frac{\text{attempt at quotient differentiation}}{\frac{1+\sin x}{\cos x}}$	M1		
		i	Fully correct differentiation	A1		
		i	Correct reduction to $\dfrac{1}{\cos x}$	A1		
		ii	Indef integral = $\ln(1 + \sin x) - \ln(\cos x)$ [Method I]	B1	or $\ln(\sec x + \tan x)$ [Method II]	
		ii	Substitute limits & use log manipulation	M1	Use of $\ln A - \ln B = \ln\dfrac{A}{B}$ anywhere in question	
		ii	Answer = $\ln(2 + \sqrt{3})$	B1	Accept ln 3.73 or $\ln\dfrac{2+\sqrt{3}}{1}$ .....but not $\ln\dfrac{1+\frac{\sqrt{3}}{2}}{\frac{1}{2}}$ Examiner's Comments This was answered surprisingly well and the logarithm manipulation was not found to be a problem. The question said "Using this result" and candidates were expected to use either the given function or some aspect of their part (i) working. If they wished to use their part (i) working and the *List of Formulae* result $\int \sec x \, dx = \ln(\sec x + \tan x)$, they were expected to quote the converse in order to obtain full marks.	Answer has not been given
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
39		i	$y=\sin^{-1}(2x)\Rightarrow\frac{dy}{dx}=\frac{1}{\sqrt{1-(2x)^2}}\cdot\frac{d(2x)}{dx}$			
		i	$=\frac{2}{\sqrt{1-4x^2}}$	B1	Oe Examiner's Comments It was surprising how many candidates could not take the formula given in the list of formulae book to get the correct derivative.	
		ii	$\frac{d^2y}{dx^2}=2\times\left(-\frac{1}{2}\right)\left(1-4x^2\right)^{-\frac{3}{2}}(-8x)=\frac{8x}{\left(1-4x^2\right)^{\frac{3}{2}}}$	B1	For correct 2nd derivative	SC 2 if result obtained correctly from $y'=\frac{k}{\sqrt{\left(1-4x^2\right)}}$
		ii	$=\frac{8x}{\left(1-4x^2\right)\sqrt{1-4x^2}}=\frac{4x}{\left(1-4x^2\right)}\frac{dy}{dx}$	M1	Using *their* ans to connect 1st and 2nd derivatives	
		ii	$\left(1-4x^2\right)\frac{d^2y}{dx^2}=4x\frac{dy}{dx}$	A1	Ft to achieve ag Examiner's Comments Notation and the layout of the question were often very poor. It was acceptable to find the left hand side and the right hand side separately and to show that they were equal. However, in a significant number of cases it was not at all clear that what was being written was one of the sides. For instance, to multiply the second derivative by (1 – $4x^2$) and show that it is equal to something else without even saying that this is $\left(1-4x^2\right)\frac{d^2y}{dx^2}$ leaves the examiner not knowing what is being done.	
		iii	$\left(1-4x^2\right)\frac{d^3y}{dx^3}-8x\frac{d^2y}{dx^2}=4\frac{dy}{dx}+4x\frac{d^2y}{dx^2}$	M1	Using result of (ii) and product rule correctly	M1 Starting with *their* 2nd derivative using appropriate method correctly

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\left(1-4x^2\right)\frac{d^3y}{dx^3}-12x\frac{d^2y}{dx^2}-4\frac{dy}{dx}=0$	A1	Examiner's Comments The more able candidates completed this part the easy way by taking the equation in (ii) and differentiating. Others took the second derivative and differentiated to find the third derivative and then attempted to substitute all three expressions into the left hand side to show that it came to 0. There were many failed attempts and a lot of fudging.	A1 ans www
		iv	Find $y_0, y'_0, y''_0, y'''_0 = \{0, 2, 0, 8\}$	B1	soi	
		iv	$y=0+2x+0+\frac{8x^3}{6}$	M1	Correctly substituting *their* 4 values into correct Maclaurin	
		iv	$\Rightarrow y=2x+\frac{4x^3}{3}$	A1	www Ignore higher order terms	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
40		i	Clear start to algebraic division	M1	at least as far as x term in quot & subseq mult back	& attempt at subtraction
		i	(Quotient) $= x - 1$	A1		
		i	(Remainder) $= x + 7$	A1		
		i	Final answer: $x-1+\frac{x+7}{x^2-x-6}$	A1	final answer in correct form This must be shown in part (i) or, if not, then implied in part (ii) If no long division shown but only comparison of coefficients or otherwise, SR M0 B1 B1 B1 Examiner's Comments This question commenced with “Use algebraic division…..” and those candidates who did not follow this instruction were penalised. In general, the division was performed well and the positions of the quotient and remainder were rarely mixed up in the final expression.	Accept $A = 1, B = -1, C = 1, D = 7$
		ii	Convert their $\frac{Cx+D}{x^2-x-6}$ to Partial Fracts	M1		
		ii	$\frac{x+7}{x^2-x-6}=\frac{2}{x-3}-\frac{1}{x+2}$ <u>Their</u> …	A1A1	Correct fraction converted to correct PFs	
		ii	$\int Ax+B\,dx=\frac{1}{2}Ax^2+Bx$ or $\frac{(Ax+B)^2}{2A}$	B1 ft		
		ii	$\int \frac{E}{x-3}+\frac{F}{x+2}\,dx=E\ln(x-3)+F\ln(x+2)$	B1 ft		
		ii	Using limits in a correct manner	M1	Tolerate some wrong signs provided intention clear	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$8+\ln\frac{27}{4}$ $\left(8+\ln\frac{54}{8}\right)$ isw	A1	Answer required in the form $a + \ln b$, so giving only a decimalised form is awarded A0 Examiner's Comments The word "Hence" was used here and candidates were expected to use their expression from part (i) and evaluate the integral from that. The use of partial fractions was necessary but was applied by only a relatively small number of candidates.	
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
41		i	$f'(x) = -\sin x.e^{-x} + \cos x.e^{-x}$	M1	Diffn using product correctly.	
		i	$\Rightarrow f'(0) = 1$	A1		
		i	$f(0) = 0$	A1	For both values www Examiner's Comments Most scored full marks on this part. Those that did not failed to differentiate a product. A few rewrote the function as the fraction $\frac{\sin x}{e^x}$ which was, of course, perfectly acceptable but gave extra opportunities for algebraic errors.	
		ii	$f'(x) = \cos x.e^{-x} - \sin x.e^{-x} = \cos x.e^{x} - f(x)$ $f''(x) = -f'(x) - \cos x.e^{-x} - f(x)$ $= -f'(x) - f'(x) - f(x) - f(x)$	M1	Diffn Examiner's Comments Most candidates confirmed that each side of the equality was $-2e^{-x}\cos x$. Having done so, a few failed to work out $f''(0)$.	
		ii	$f''(x) = -2f'(x) - 2f(x)$ OR $-2\cos x.e^{-x}$	A1		
		ii	Showing the two equal	A1		
		ii	$f''(0) = -2$	A1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$f''(x) = -2f'(x) - 2f(x)$ $\Rightarrow f'''(x) = -2f''(x) - 2f'(x)$ oe	B1	Not involving trig or exp fns Examiner's Comments A significant number only found $f'''(0)$. A few others made a guess as to what $f'''(x)$ might look like. [$-2f''(x) - 2f'(x) - 2f(x)$ was the most popular, closely followed by $-3f''(x) - 3f'(x)$.] Most candidates failed to realise that all that was required was to differentiate $f''(x)$ from part (ii).	$= -f' + 2f$ Or $2f' + 4f$
		iii	$\Rightarrow f'''(0) = 4 - 2 = 2$	B1		
		iv	$f(x) = x - x^2 + \frac{x^3}{3}$	M1		
		iv		A1		
		iv	Alternative: Write down correct series expansion for e^{-x} and $\sin x$ and multiply	M1		
		iv		A1	Examiner's Comments Most obtained the method mark for using the Maclaurin series, albeit with incorrect values from the earlier parts. Just a few candidates ignored all the work in the first three parts and wrote down, and multiplied, the series expansions for $\sin x$ and e^x from the *List of Formulae*. This was, of course, acceptable but added time.	
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
42			$\int_0^2 \frac{1}{\sqrt{4+x^2}}\,dx = \left[\sinh^{-1}\left(\frac{x}{2}\right)\right]_0^2$	M1	Standard form	
			$= \sinh^{-1} 1 - \sinh^{-1} 0$	M1	Use of log form and substitute limits dep on 1st M	
			$= \ln\left(1+\sqrt{1+1}\right)-0$	A1		
			$= \ln\left(1+\sqrt{2}\right)$ **cao** **isw**			
			Alternative: $\int_0^2 \frac{1}{\sqrt{4+x^2}}\,dx = \left[\ln\left(x+\sqrt{x^2+4}\right)\right]_0^2$	M1	Standard form	
				M1	Substitute limits Examiner's Comments This was usually done well and provided a solid start to the paper. It was expected that candidates would give their final answer as a single logarithm.	
			$= \ln\left(2+\sqrt{8}\right)-\ln 2$			
			$= \ln\left(1+\sqrt{2}\right)$	A1		
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
43		i	$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$	B1	2 or 3 terms correct unsimplified	
		i	$\Rightarrow \ln(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \frac{x^8}{4} \ldots$	B1	All terms correct	
		i	Validity: $-1 \le x \le 1$ or $\lvert x \rvert \le 1$	B1		
		ii	$\ln(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \frac{x^8}{4} - \ldots$			
		ii	Substitute $x = \frac{1}{2}$	M1	Sub $x = \frac{1}{2}$ into *their* ans to (i)	Alt: divide by x^2 then sub
		ii	$\Rightarrow \ln\left(\frac{5}{4}\right) = \left(\frac{1}{2}\right)^2 - \frac{1}{2}\left(\frac{1}{2}\right)^4 + \frac{1}{3}\left(\frac{1}{2}\right)^6 - \frac{1}{4}\left(\frac{1}{2}\right)^8 + \ldots$ $= \frac{1}{4}\left(1 - \frac{1}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{3}\left(\frac{1}{2}\right)^4 - \frac{1}{4}\left(\frac{1}{2}\right)^6 + \ldots\right)$			
		ii	$\Rightarrow \left(1 - \frac{1}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{3}\left(\frac{1}{2}\right)^4 - \frac{1}{4}\left(\frac{1}{2}\right)^6 + \ldots\right)$ $= 4\ln\left(\frac{5}{4}\right)$ **isw**	A1	Single ln expression Examiner's Comments Most candidates were also able to write down the series expansion. The range of values of x for which the series is valid was, however, poorly done and 3 marks was the norm.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
44		i	Integrate to obtain $k(4x+1)^{\frac{1}{2}}$ or $ku^{\frac{1}{2}}$	*M1	any constant k	
		i	Obtain correct $\frac{1}{2}\sqrt{3}(4x+1)^{\frac{1}{2}}$ or $\frac{1}{2}\sqrt{3}u^{\frac{1}{2}}$	A1	or exact equiv	
		i	Apply limits 0 and 20 and attempt subtraction of area of rectangle (or limits 1 and 81 if u involved)	M1	dep *M; or equiv such as including term $-\frac{1}{9}\sqrt{3}$ in the integration or finding $\int\frac{1}{9}\sqrt{3}$ separately; allow M1 if decimal values used here	Alternative: (region between curve and y-axis) Obtain equation $x=\frac{3}{4}y^{-2}-\frac{1}{4}$ B1 Integrate to obtain form $k_1 y^{-1} + k_2 y$ *M1 Apply limits $\frac{1}{9}\sqrt{3}$ and $\sqrt{3}$ the right way round M1 d*M
		i	Obtain $4\sqrt{3}-\frac{20}{9}\sqrt{3}$ and hence $\frac{16}{9}\sqrt{3}$	A1	answer must be exact and a single term; $\frac{16}{9}\sqrt{3}+c$ as answer is final A0	Obtain $\frac{6}{\sqrt{3}}-\frac{8}{36}\sqrt{3}$ or better A1
		ii	State volume is $\pi\int\frac{3}{4x+1}\,dx$	B1	no need for limits here; condone absence of dx; condone absence of π here if it appears later in solution	allow B1 for $\int\pi y^2$ and $y^2=\frac{3}{4x+1}$ stated
		ii	Obtain integral of form $k\ln(4x+1)$	M1	any constant k with or without π	if brackets missing, and subsequent calculation does not show their 'presence', marks are max B1M1A0A0M1A0
		ii	Obtain $\frac{3}{4}\pi\ln(4x+1)$ or $\frac{3}{4}\ln(4x+1)$	A1		
		ii	Apply limits to obtain $\frac{3}{4}\pi\ln 81$ or $\frac{3}{4}\ln 81$	A1	or exact equiv perhaps with ln1 present	
		ii	Attempt to subtract volume of cylinder, using correct radius and 'height'	M1	with exact volume of cylinder attempted	do not treat rotation around y-axis as mis-read: this is 0/6

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Obtain $3\pi\ln 3-\frac{20}{27}\pi$ or $\pi(\frac{3}{4}\ln 81-\frac{20}{27})$ and attempts leading to $\ln(4x+1)$. For c	A1	or exact equiv involving two terms Examiner's Comments There were more challenges for candidates in the two parts of this question. There were inaccuracies in carrying out the integration and in using the correct limits. A significant number of candidates ignored the need to deal with the region between R and the x-axis and produced answers that applied to a region between the curve and both axes. Common errors in part (i) included initial steps such as $\int 3(4x+1)^{\frac{1}{2}}\,dx$ and $\int 3(4x+1)^{-\frac{1}{2}}\,dx$. Errors in the integration included $\int(\frac{3}{4x+1})^{\frac{1}{2}}\,dx$ leading to an expression involving ndidates with the correct $\int\sqrt{3}(4x+1)^{-\frac{1}{2}}\,dx$, there was no certainty that the correct $\frac{1}{2}\sqrt{3}(4x+1)^{\frac{1}{2}}$ would follow. Incorrect limits such as	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					$\frac{1}{9}\sqrt{3}$ and $\sqrt{3}$ or $\frac{1}{9}\sqrt{3}$ and 20 were sometimes used. An alternative approach was used by a number of candidates who treated *R* as a region between the curve and the *y*-axis. This needed care in expressing *x* in terms of *y*, in carrying out the integration accurately and in choosing the correct limits but many adopting this approach did reach the correct answer. They were among the 49% of candidates who did earn full marks for part (i). Some of the candidates treating *R* as a region between the curve and *y* -axis in part (i) continued in part (ii) to attempt to find the volume of revolution round the *y*-axis. Whether this was just careless reading of the question or whether they thought that the answer would be the same whichever axis they used was not clear. Generally in part (ii) candidates seemed to be on more familiar ground with the integration and most recognised that the result involved a natural logarithm although the integration often led to the incorrect $3\pi\ln(4x + 1)$. Again, incorrect limits were applied in many cases. Many candidates did nothing about removing the volume of the unwanted cylinder; did they forget or did they not appreciate that this extra step was necessary? Of those making an attempt at this	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					final step, many merely subtracted $\frac{20}{9}\sqrt{3}$ as they had in part (i) and others seemed to treat the shape to be removed as a cuboid or circle. 40% of the candidates managed to record all six marks on this part.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
45		i	$x = \cosh y = \frac{e^y + e^{-y}}{2} \Rightarrow e^y + e^{-y} = 2x$ $\Rightarrow e^{2y} - 2xe^y + 1 = 0$	M1	Finding 3 term quadratic in e^y	
		i	$\Rightarrow e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}$ $\Rightarrow y = \ln\left(x \pm \sqrt{x^2 - 1}\right)$	A1	Correct solution	Condone ignoring −ve sign at this point.
		i	Reject – sign as principal value taken	B1	Including reason oe Examiner's Comments Most candidates set up a quadratic equation in e^y and solved it to find e^y and hence y. However, only a few candidates convincingly explained why the positive root needed to be taken, often referring in a vague way to the expression needing to be positive. The best explanations were those that talked about the shape of the graph of $y = \cosh^{-1} x$.	Condone interchange of x and y but final ans must be correct
		i	$\Rightarrow y = \ln\left(x + \sqrt{x^2 - 1}\right)$	A1		
		ii			Alt:	
		ii	$y = \ln\left(x + \sqrt{x^2 - 1}\right) \Rightarrow \frac{dy}{dx} = \frac{1}{\left(x + \sqrt{x^2 - 1}\right)} \times \left(1 + \frac{2x}{2\sqrt{x^2 - 1}}\right)$	M1	$x = \cosh y \Rightarrow \frac{dy}{dx} = \frac{1}{\sinh y}$	
		ii	$= \frac{1}{\left(x + \sqrt{x^2 - 1}\right)} \times \frac{\left(x + \sqrt{x^2 - 1}\right)}{\sqrt{x^2 - 1}} = \frac{1}{\sqrt{x^2 - 1}}$	A1	$= \frac{1}{\sqrt{x^2 - 1}}$ Examiner's Comments Was often answered well, using the derivative of the inverse ($x = \cosh y$) tended to create fewer algebraic errors than direct differentiation.	
		iii	$x = \cosh^{-1} 3$	M1	Use of $\cosh^{-1}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= \ln\left(3+\sqrt{8}\right)$	A1		
		iii	$= -\ln\left(3+\sqrt{8}\right)$ **oe**	A1	ft, –ve the first answer Examiner's Comments Several candidates lost the second solution.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
46			$\cos\theta = \dfrac{1-t^2}{1+t^2}$	M1	Using t substitution for both $\cos\theta$ and $\mathrm{d}\theta$	
			$\dfrac{\mathrm{d}t}{\mathrm{d}\theta} = \dfrac{1}{2}\sec^2\dfrac{1}{2}\theta = \dfrac{1}{2}\left(1+\tan^2\dfrac{1}{2}\theta\right)$	A1	Subs correct	
			$\Rightarrow \quad \mathrm{d}t = \dfrac{1+t^2}{2}.\,\mathrm{d}\theta \Rightarrow \mathrm{d}\theta = \dfrac{2\mathrm{d}t}{1+t^2}$	M1	Dealing with limits and attempting integration.	
			$\Rightarrow I = \displaystyle\int_0^1 \frac{1}{1+\frac{1-t^2}{1+t^2}}\,\frac{2\mathrm{d}t}{1+t^2} = \int_0^1 \frac{1+t^2}{1+t^2+1-t^2}\,\frac{2\mathrm{d}t}{1+t^2}$	A1	Correct integral	
			$\displaystyle\int_0^1 \frac{2\mathrm{d}t}{2} = [t]_0^1 = 1$	A1	Answer Examiner's Comments There were many good solutions to this first question and a variety of methods for obtaining the correct answer. Many candidates did not include the “dθ” in their integral and, as a result, failed to complete the full substitution. Some failed to convert the limits and so gave an incorrect answer. Others converted back unnecessarily.	
			Alternative $1+\cos\theta = 2\cos^2\frac{1}{2}\theta$ $\Rightarrow \displaystyle\int_0^{\frac{\pi}{2}} \frac{1}{1+\cos\theta}\mathrm{d}\theta = \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{1}{\cos^2\frac{1}{2}\theta}\mathrm{d}\theta = \frac{1}{2}\int_0^{\frac{\pi}{2}} \sec^2\tfrac{1}{2}\theta\,\mathrm{d}\theta$ $= \dfrac{1}{2}\left[2\tan\dfrac{1}{2}\theta\right]_0^{\frac{\pi}{2}} = \tan\dfrac{\pi}{2} - \tan 0 = 1$	SC3		
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
47		i	Integrate to obtain form $k(3x+1)^{\frac{1}{2}}$	*M1	any non-zero constant k	
		i	Obtain $4(3x+1)^{\frac{1}{2}}$	A1	or (unsimplified) equiv; or $4u^{\frac{1}{2}}$ following substitution	
		i	Apply the limits and subtract the right way round	M1	dep *M	
		i	Obtain $4\sqrt{28}-4\sqrt{7}$ and show at least one intermediate step in confirming $4\sqrt{7}$	A1	AG; necessary detail required; decimal verification is A0; $[\ldots]_2^9 = 4\sqrt{28}-4\sqrt{7}=4\sqrt{7}$ is A0; $[\ldots]_2^9 = 8\sqrt{7}-4\sqrt{7}=4\sqrt{7}$ is A0 <u>Examiner's Comments</u> 50% of the candidates recorded all four marks and another 34% recorded three marks out of four. These figures reflect the fact that most of the candidates had no difficulty with the integration although a few did rewrite y as $(3x+1)^{\frac{1}{2}}$. The main problem concerned the fact that the answer was given in the question and that, as a result, candidates were expected to provide a convincing conclusion. Candidates going immediately from $4\sqrt{28}-4\sqrt{7}$ to the given $4\sqrt{7}$ did not earn the final mark. An extra step or two to justify the conclusion was needed. Use of a calculator to compare decimal approximations of $4\sqrt{28}-4\sqrt{7}$ and $4\sqrt{7}$ also did not earn the final mark.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	State or imply volume is $\int \pi \left(\frac{6}{\sqrt{3x+1}}\right)^2 dx$ or equiv	B1	merely stating $\int \pi y^2\, dx$ not enough; condone absence of dx; no need for limits yet; π may be implied by its later appearance	
		ii	Integrate to obtain $k \ln(3x + 1)$	M1	any non-zero constant with or without π	
		ii	Obtain $12\pi \ln(3x + 1)$ or $12 \ln(3x + 1)$	A1	or unsimplified equiv	
		ii	Substitute limits correct way round and show each logarithm property correctly applied	M1	allowing correct applications to incorrect result of integration providing natural logarithm involved; evidence of $\ln 28 - \ln 7 = \frac{\ln 28}{\ln 7}$ error means M0	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Obtain $24\pi \ln 2$	A1	no need for explicit statement of value of k Examiner's Comments Candidates knew in general terms what was required in this question but the details were often incorrect. There were a few slips in squaring y and π was sometimes missing but the vast majority started with $\int \frac{36\pi}{3x+1} dx$. Integrating to obtain $36\pi \ln(3x$ $+ 1)$ was a common mistake. Some careful work involving logarithm properties was then needed to present the answer in the required form and many candidates struggled at this point. Most were able to reach an expression involving ln4 but concluding with an answer in the form k ln 2 unexpectedly proved to be challenging for some. Some stopped as soon as they reached $12\pi \ln 4$ and, for others, division throughout by 2 was their means of moving from 12π ln 4 to $6\pi \ln 2$. For a routine request, it is a little disappointing to note that only 41% of the candidates recorded full marks.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
48		i	$\frac{dy}{dx}=\frac{1}{1-\left(\frac{1-x}{3+x}\right)^2}\times\frac{-(3+x)-(1-x)}{(3+x)^2}$	B1	Sight of standard diffn for $\tanh^{-1}x$	
		i		M1	Fn of fn and diffn of quotient	
		i		A1	Soi correct quotient (i.e. correct expression for 2nd part)	
		i	$\Rightarrow\frac{dy}{dx}=\left(\frac{-4}{(3+x)^2-(1-x)^2}\right)=\frac{k}{1+x}$	A1		
		i	$\Rightarrow\frac{dy}{dx}=\frac{-1}{2(1+x)}$	A1	Correct for y'	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$\Rightarrow \frac{d^2y}{dx^2} = \frac{1}{2(1+x)^2}$	A1	2^{nd} diffn (NB AG) Examiner's Comments A significant proportion failed to use the chain rule in the original differentiation and consequently made little progress. Many candidates who made this error did not understand that what they were doing was not going to produce the result required and was involving them in very complicated algebra. Other common errors included not writing down the derivative of $\tanh^{-1} x$ correctly, even though it is in the formula book. In order to obtain the correct result for f″(x) it was necessary to find f′(*x*). There were some who followed an incorrect route but still came up with the correct f′(*x*), presumably by working backwards from the given f″(*x*). This, of course, was not creditworthy. An alternative method was to rewrite f(*x*) in terms of natural logarithms. For those who could cope with the algebra this often produced a shorter and neater solution.	
		ii	When $x = 0, y = \tanh^{-1}\frac{1}{3}$ or $\frac{1}{2}\ln 2$ or $\ln\sqrt{2}$ $\frac{dy}{dx} = -\frac{1}{2}$	B1	For 1^{st} value (needs to be exact)	
		ii	$\frac{d^2y}{dx^2} = \frac{1}{2}$	B1	For both	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow y = \tanh^{-1}\frac{1}{3} + \left(-\frac{1}{2}\right)x + \left(\frac{1}{2}\right)\frac{x^2}{2}$	M1	Use of correct Maclaurin's series	
		ii	$= \tanh^{-1}\frac{1}{3} - \frac{1}{2}x + \frac{x^2}{4}$	A1	Accept 0.347 Examiner's Comments Those candidates who wrote down the general form of Maclaurin's expansion first, rarely omitted the 2 from the denominator of the third term. Several candidates who were unsuccessful in proving the results in part (i) worked backwards from the given result and were able to pick up the marks in this part.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
49		i	Width of rectangles is $\frac{3}{n}$	B1	Statement about width	
		i	$\Rightarrow$ Sum of areas of rectangles	M1	Height or area of at least one rectangle	
		i	$=\frac{3}{n}\times\left(\ln(\ln 3)+\ln\left(\ln\left(3+\frac{3}{n}\right)\right)+\ldots\ldots\right)$	A1	Correct conclusion www	
		i	$=\frac{3}{n}\times\sum_{r=0}^{n-1}\ln\left(\ln\left(3+\frac{3r}{n}\right)\right)$		1468 or Examiner's Comments Many seemed to be only expanding the given summation. Examiners needed to be convinced that what was being written down was not just the given answer. At the very least, candidates were expected to describe the areas as being the width multiplied by the height for each rectangle. A diagram usually helped. Many candidates who had problems here then gave up on the rest of the question.	
		ii	$=\frac{3}{n}\times\sum_{r=1}^{n}\ln\left(\ln\left(3+\frac{3r}{n}\right)\right)$	B1	Examiner's Comments This part was nearly always correct. An error for some was to leave the lower limit r = 0 alone, thus creating an extra rectangle.	
		iii	$U-L=\frac{3}{n}\times\ln(\ln 6)-\frac{3}{n}\times\ln(\ln 3)$	M1*	Subtraction to obtain the difference of two terms	
		iii	$=\frac{3}{n}(\ln(\ln 6)-\ln(\ln 3))=\frac{3}{n}\ln\left(\frac{\ln 6}{\ln 3}\right)$	A1		
		iii	$\Rightarrow n>\frac{3}{0.001}\ln\left(\frac{\ln 6}{\ln 3}\right)\Rightarrow n>\frac{3}{0.001}\times\ln(1.6309)$	*M1	Dealing with inequality to obtain n dep on first M	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	⇒ least $n = 1468$	A1	Accept $n \geq 1468$ or $n > 1467$ Examiner's Comments There were many good attempts but plenty of opportunity for slips. The most common misunderstanding was that $U - L$ gave a single term, usually ln(ln6). Common slips included ln(ln…) becoming just ln… or totally disappearing, the difference being written the wrong way round, the right-hand side appearing as 0.01 or the final answer as being given as $n > 1468$.	
			Total	8		